HRI: Harmonizing Rotation Invariance for Remote Sensing Pretraining Model

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***Abstract*—In remote sensing, self-supervised pretraining methods like masked image modeling (MIM) and contrastive learning (CL) have demonstrated promising image understanding capabilities. However, existing approaches largely overlook the critical challenge of rotation diversity, failing to learn robust rotation-invariant representations. To address this gap, particularly given the arbitrary orientations of ground targets and imaging angles, we propose a novel pretraining framework hierarchical rotation-invariant (HRI) that systematically integrates rotation-aware tasks, enabling the model to learn rotation-invariant features from multiple levels while preserving semantic consistency across rotational transformations. Specifically, we introduce a masked multiangle reconstruction strategy that, for the first time, reconstructs masked images at various rotated angles to capture pixel-level rotation-invariant and textural features. In addition, we exploit the rotation-averaged features for CL, which helps suppress noise and enhances the learning of robust, global rotation-invariant representations. Extensive experiments across a range of downstream tasks demonstrate that our method outperforms existing approaches in robustness and accuracy under varying rotation conditions. The code will be released at https://github.com/mint0126/HRI**

***Index Terms*—Remote sensing, rotation invariance, self-supervised pretraining.**

I. Introduction

REMOTE sensing images serve as critical resources in environmental monitoring [1], urban planning [2], agricultural assessment [3], and disaster management [4]. Recent advancements in satellite and drone platforms have exponentially expanded the volume of remotely sensed data, enabling large-scale geospatial analysis. However, the inherent complexity of these images, characterized by cluttered backgrounds and dense multiobject scenes, makes manual annotation cumbersome and costly. To address this, researchers have developed self-supervised learning (SSL) paradigms, such as masked image modeling (MIM) [5], [6] and contrastive learning (CL) [7], [8], which reduce annotation dependence by leveraging pretext tasks to learn discriminative low-level features and hierarchical semantic representations. These approaches have demonstrated significant performance improvements in downstream tasks, including object detection [9], [10], fine-grained classification [11], [12], and pixel-level segmentation [13], [14].



Fig. 1. Examples of different ground objects with various directions.

As shown in Fig. 1, remote sensing images inherently capture an overhead perspective, where objects exhibit arbitrary orientations, unlike natural images, where objects typically adhere to canonical orientations (e.g., upright faces). Existing self-supervised pretraining methods overlook the critical impact of rotational variability on feature extraction, resulting in suboptimal robustness when processing rotated instances. As demonstrated in Table I, the random rotation of input images causes significant performance degradation across tasks like classification and segmentation, underscoring the imperative to learn rotation-invariant representations. Such invariance is essential for reliably interpreting terrain characteristics across arbitrary viewing angles, a prerequisite for robust geospatial analysis. Rotation-based pretraining has

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TABLE I
QUANTIFICATION OF ROTATION INDUCED PERFORMANCE CHANGES IN CLASSIFICATION ACCURACY (%) AND SEMANTIC SEGMENTATION MIOU (%), EVALUATED BY LINEAR PROBING WITH AND WITHOUT RANDOM ROTATIONS

Task	Dataset	Model	W/O Rot	With Rot	Δ
Classification	UCM	CMID [21]	92.37	89.42	-2.95
		Ours	94.53	93.15	-1.38
Segmentation	Vaihingen	CMID [21]	80.80	79.11	-1.69
		Ours	82.21	81.40	-0.81

emerged as a pivotal strategy in SSL. RotNet [15] pioneered this paradigm by introducing a rotation prediction task where model learns to infer the rotation angles (0°, 90°, 180°, 270°) applied to input images, driving the model to perceive rotational transformations, thereby learning discriminative feature representations with enhanced robustness to geometric variations. Studies in natural image domains [16], [17], [18] demonstrate that integrating rotation priors could enhance the model's robustness to viewpoint variations. In the remote sensing field, where overhead perspectives inherently exhibit arbitrary object orientations. Preliminary work [19], [20] in this field extracts rotation-related information, bridging the gap between geometric invariance and semantic discriminability in complex aerial scenes.

While the geospatial structures in remote sensing imagery, such as farmland textures and road networks, often exhibit rotational diversity, their semantic and structural properties remain consistent across arbitrary orientations. Therefore, rotation-invariant pretraining holds great promise. By pretraining models to disentangle rotational transformations from semantic content, such methods could improve the adaptability to multidirectional ground objects, which is critical for tasks like multiangle object detection and scene classification in remote sensing. To exploit this characteristic, we propose a rotation-based multitask self-supervised pretraining framework that synergistically combines masked multiangle reconstruction, masked rotation-averaged CL, and masked local semantic alignment.

Masked multiangle reconstruction task requires the model to reconstruct masked images not only into the original input (0°) but also into images rotated at 90°, 180°, and 270°. It adds a learnable token to the masked patches to learn the contextual information during feature extraction. Then, the four reconstructed images with different angles are obtained by four lightweight decoders. Different from previous work, the proposed framework pioneers multiangle reconstruction from a single masked image, compelling the model to learn spatial projection patterns of masked regions across varying rotation angles. It achieves disentanglement of content and orientation by establishing a rotation-spatial mapping mechanism, and enables cross-view semantic alignment through learned rotation-invariant representations, ensuring consistency in reconstructed semantics despite geometric transformations.

Masked rotation-averaged CL exploits the unmasked averaged features calculated by four random angles and the masked image features from four random angles based on MoCo [22] to achieve cooperative enhancement of rotation invariance and occlusion robustness. Rotation-averaged contrast learning forces the model to disentangle rotation transformations from semantic content and focus on rotation-invariant features, while masked CL forces the model to recover the occluded area by using context and enhance cross-view consistency. In addition, averaging features from four rotation angles could suppress noise introduced by rotation while preserving essential semantics of rotation invariance.

Masked local semantic alignment aims to minimize the distribution difference between the masked features from the student (masked) branch and the augmented features from the teacher (unmasked) branch. It first matches the masked regions and the cropped-augmented regions to align them at the locations, then calculates the prototypes of the masked and augmented images to obtain the similarity distributions. This task forces the model to focus on the semantic consistency between the masked regions and the augmented image by aligning the prototypes' semantics between position-matched pairs and mitigate the loss of dense small targets in remote sensing images due to the masking operation.

These three tasks collectively enhance the model's feature extraction capability for complex remote sensing scenarios through: 1) hierarchical rotation-invariant (HRI) feature learning; 2) multitask loss combination; and 3) the stable training mechanism of the MoCo framework. Together, they establish a robust pretraining foundation for downstream tasks. Notably, our masked rotation-averaged CL specifically addresses directional sensitivity by effectively extracting rotation-invariant high-level semantic features.

In summary, our main contributions are as follows.

1) We introduce HRI learning, an innovative self-supervised framework that jointly models multilevel rotation-invariant representations to enhance feature robustness for remote sensing imagery.
2) We design two complementary rotation-based tasks to enhance feature representation: a) a multiangle reconstruction task that enforces spatial alignment of contextual features across orientations at the pixel-level, enabling the recovery of occluded structures and reasoning about directional dependencies; and b) a rotation-average CL task that aligns semantic representations of rotated views at image-level, promoting rotation invariance while distinguishing unrelated features.
3) Our proposed method demonstrates outstanding performance across diverse downstream tasks, including scene classification, rotated object detection, semantic segmentation, and change detection, demonstrating its versatility in addressing complex challenges inherent to remote sensing imagery, such as arbitrary orientations, occlusions, and dynamic environmental conditions.

The rest of this article is organized as follows. Section II reviews the related work. Section III details the proposed pretraining framework. Section IV provides extensive experiments and benchmark evaluations, followed by the conclusion in Section VI.

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II. Related Work

A. Self-Supervised Pretraining for Remote Sensing

Numerous RS-specific pretrained models have proliferated to address the unique challenges of RS image analysis. Among these, SSL has emerged as a critical research direction, enabling models to harness vast unlabeled RS datasets for pretraining while mitigating the challenges of costly manual annotation and limited labeled data availability. Current SSL frameworks for RS can be broadly categorized into three paradigms based on their pretraining strategies: 1) MIM-based approaches [5], [6], [23], [24], [25], which learn contextual feature recovery; 2) CL-based methods [7], [8], [26], which optimize feature invariance across augmentations; and 3) multitask learning-based frameworks [21], [27], [28], which unify complementary self-supervised objectives.

MIM randomly masks partial regions of an image and reconstructs the masked parts via the unmasked regions, enabling the model to obtain pixel-level image understanding. Scale-MAE [6] encodes the positional encoding via the area of the Earth covered by the image rather than the image resolution to obtain scale-specific information and decodes the masked image through a bandpass filter to reconstruct low-/high-frequency images at lower/higher scales. RingMo [5] introduces a patch incomplete mask (PIMask) strategy tailored for remote sensing imagery, addressing the challenges posed by dense and small objects in such scenes. Wang et al. [23] introduce a rotated variable-size window attention mechanism to replace standard full attention in transformers, significantly reducing computational costs while effectively capturing object scale and directional information. FG-MAE [24] introduces a novel reconstruction objective that combines histograms of oriented gradients (HOGs) with normalized difference indices (NDIs) for SSL on multispectral imagery. MURLF [25] proposes a multimodal MIM framework that employs independent multilevel masking across modalities, while leveraging both intramodal and cross-modal visual tokens to reconstruct masked pixels as its pretext task.

CL achieves SSL by constructing positive and negative samples and then measuring the distance between the positive and negative samples, with a primary focus on image-level feature understanding. Ayush et al. [7] propose to leverage spatially aligned images over time to construct temporal positive pairs in CL and utilize the geo-location in the design of pretext tasks. PIS [8] emphasizes the promotion of intrainstance similarity through a modified CL framework, which capitalizes on the distinct attributes of RSIs and harnesses a wider range of potential variations to discover the characteristics of intrainstance, thereby extracting more resilient representations for downstream tasks. Jiang et al. [26] apply active learning methods to select subsets for self-supervised pretraining based on CL to improve training efficiency while reducing labeling costs.

Multitask learning has emerged as a powerful strategy for pretraining robust and generalizable feature representations by unifying complementary self-supervised objectives. The combination of CL and MIM could balance the understanding of global semantics and local spatial relationships. Park et al. [27] theoretically validate the effectiveness of combining MIM and CL, where CL primarily captures global dependencies, while the self-attention mechanism in MIM mainly captures local relationships. CMAE [28] consists of an asymmetric encoder–decoder as an online branch and a momentum-updated encoder as a target branch. It introduces pixel shifting and feature decoding to make CL compatible with MIM and complement the contrastive features, respectively. CMID [21] proposes a unified SSL framework that is capable of learning representations with both global semantic separability and local spatial perceptibility by combining CL with MIM in a self-distillation way.

Our proposed method belongs to the multitask SSL framework, which considers the issue of various angles of targets in remote sensing images, which has been ignored in previous works. We exploit the rotation-averaged CL and the modified MIM to extract rotation-invariant and robust features.

B. Rotation-Based Task

Rotation prediction is one of the representative rotation-based pretext tasks defined in SSL, which has been widely used in the natural images field. RotNet [15] proposes a simple rotation prediction task for pretraining, providing a powerful supervisory signal for learning semantic features. E-Rot [16] jointly predicts rotation angles and image enhancement transformations, enabling the model to learn both object shape and texture representations by analyzing appearance modifications. MyDLNote-Network [17] decouples the representations by using a rotation prediction task and an instance recognition task, thus learning both rotation-invariant and rotation-sensitive features. Zheng et al. [18] introduce a regularization loss based on kernel mean embeddings with rotation-invariant kernels on the hypersphere for SSL of image representations. Xu et al. [29] exploit rotation angle prediction for semi-supervised learning and propose conditional rotation angle estimation (CRAES) by adopting a module that predicts the rotation angle of the image conditioned on the candidate image class. Lee et al. [30] leverage group equivariant CNNs to extract discriminative rotation-invariant descriptors and obtain rotation equivariant features and their orientations explicitly without having to perform sophisticated data augmentations.

Rotation-based tasks have also been used in different remote sensing tasks, i.e., semantic segmentation and image retrieval. Zhao et al. [19] propose a rotation-aware building instance segmentation network that integrates a refined rotated detector to extract rotation equivariant and invariant features. Cai et al. [31] design a proxy-based Euclidean distance contrast loss that combines contrast learning within the framework of metric learning. It endows the model with a certain degree of rotation invariance by pulling the distance between the source image and its rotated image embedding vector in the metric space closer than any other image. Wen et al. [32] propose a novel rotation awareness-based learning framework for SAR target recognition with limited training samples by exploiting intrinsic relation constraints among these sequences as supervision to predict the rotational pattern of a baseline sequence. Li et al. [33] introduce a scaling center crop operation to generate

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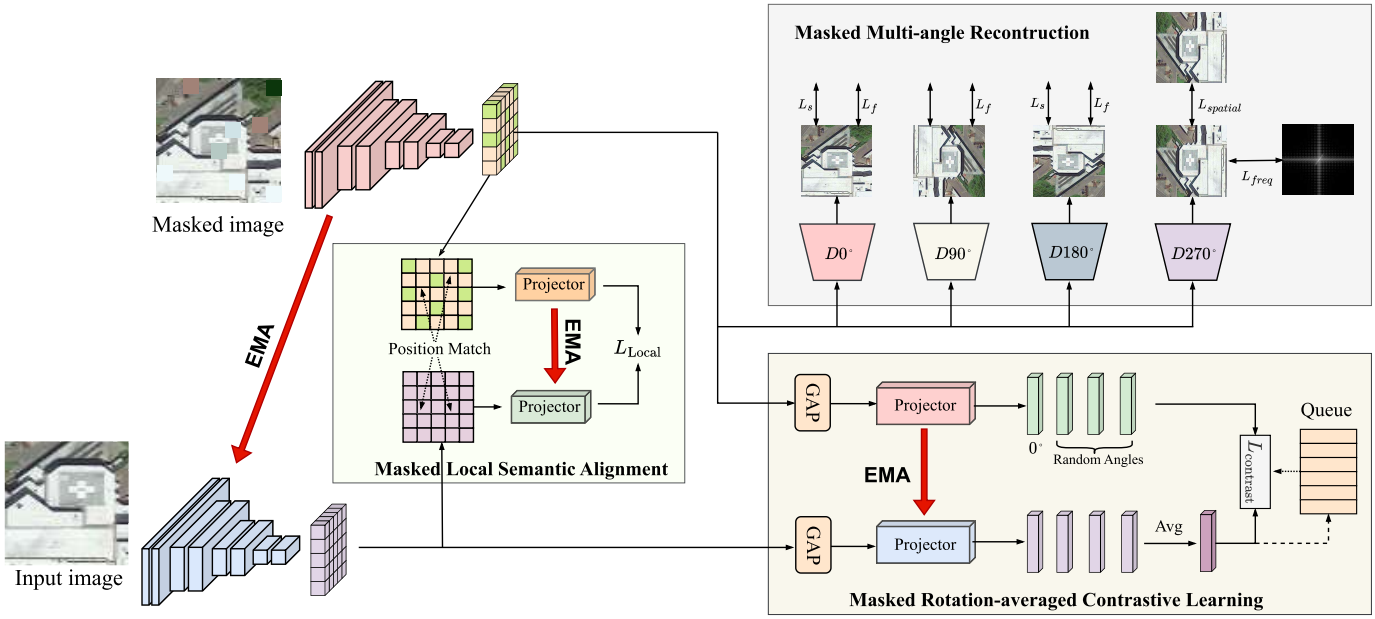


Fig. 2. Framework of our SSL method. The visual encoder is trained with three proxy tasks: masked local semantic alignment, masked multiangle reconstruction, and masked rotation CL.

diverse rotated crops, then learns rotation-invariant representations by reconstructing these transformed views back to the original images.

Different from the existing methods, we propose two rotation-based tasks to learn low-level and high-level rotation-invariant features, and propose to reconstruct multiple-angle images from a single input masked image for the first time.

III. Method

A. Overview

As shown in Fig. 2, our method employs a dual-branch framework—one for unmasked images and another for masked images, using multitask SSL to extract rotation-invariant and generalizable representations from unlabeled remote sensing data. The model integrates three complementary tasks.

1) Masked multiangle reconstruction reconstructs the full content of the input image across all four rotation angles from a single masked input via four decoders, thereby enhancing context modeling and perceiving low-level rotation-invariant features.
2) Masked rotation-averaged CL promotes rotation-invariant representations by aligning the rotation-average features across masked images from different rotation angles based on the dynamic queue and momentum update from MoCo [22], thereby obtaining high-level rotation-invariant features.
3) Masked local semantic alignment enforces consistency between position-matched patches from the unmasked and masked models, alleviating the semantic incompleteness in masked regions and improving the local semantic consistency between masked and unmasked features.

By jointly optimizing these tasks, the model learns hierarchical structured features that balance rotation invariance with task-sensitive discriminability. This dual-property representation enables strong adaptability to diverse downstream remote sensing applications, including land cover classification, object detection, semantic segmentation, and change detection. Below, we detail the three self-supervised tasks.

B. Masked Multiangle Reconstruction

Unlike MAE, which processes only unmasked patches, our method fills masked patches with their mean spectral values while incorporating learnable tokens, enabling global context capture during feature extraction through these adaptive masked-region tokens. Considering the maximum geometric diversity of object appearance directions in remote sensing images, we propose to restore the image to a reconstructed image from four angles (0°, 90°, 180°, 270°) via four lightweight decoders for four different directions modeling.

The input image is first divided into nonoverlapping patches x^p. Following [21], we apply a blockwise masking strategy with mask ratio r and fill masked regions with the mean spectral value to preserve small, dense objects that are prevalent in remote sensing imagery. To maintain contextual learning, we inject learnable [MASK] tokens into masked patches, enabling the model to leverage unmapped regions for feature extraction. Formally

$$\mathrm{x}^{\mathrm{mask}} = \mathrm{x}^p + [\mathrm{MASK}] \cdot \mathbf{M} \tag{1}$$

where $\mathrm{x}^{\mathrm{mask}}$ is the processed masked images, and $\mathbf{M}$ is the binary masking matrix.

Then, the encoder E encodes the masked image $\mathrm{x}^{\mathrm{mask}}$ to produce latent feature embeddings $\mathbf{F}^{\mathrm{mask}}$. These embeddings are then processed by four lightweight decoders to reconstruct the image at four rotation angles {0°, 90°, 180°, 270°}, and each decoder consists of a 3 × 3 convolutional layer for

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reconstruction followed by a PixelShuffle operation for upsampling

$$\mathbf{F}^{\text{mask}} = E\left(\mathrm{x}^{\text{mask}}\right) \tag{2}$$

$$\hat{\mathrm{x}}_\theta = D_\theta\left(\mathbf{F}^{\text{mask}}\right), \quad \theta \in (0^\circ, 90^\circ, 180^\circ, 270^\circ) \tag{3}$$

where D_θ is the reconstruction decoder for rotation angle θ.

The reconstruction loss combines both spatial and frequency domain constraints. We employ an ℓ_1 loss for spatial reconstruction and an ℓ_2 loss (mse) for frequency domain alignment

$$L_{\text{reconstruct}} = \ell_1\left(\hat{\mathrm{x}}_\theta, \mathrm{x}_\theta\right) + \ell_2\left(\text{FFT}\left(\hat{\mathrm{x}}_\theta\right), \text{FFT}\left(\mathrm{x}_\theta\right)\right) \tag{4}$$

where FFT(·) denotes the fast Fourier transform.

The masking operation forces the model to infer missing regions using global and contextual spatial dependencies, and the multiangle reconstruction requires the model to preserve both content and geometric relationships (e.g., ensuring a rotated building maintains its structural integrity). Jointly optimizing reconstruction across angles ensures features encode disentangled representations of content and orientation, promoting contextual reasoning and rotation-invariant understanding.

C. Rotation-Averaged CL

Rotation transformations serve as an effective data augmentation strategy, as different rotated views of the same image exhibit higher feature similarity compared to distinct images. To leverage this property, we employ MoCo-style CL [22] to simultaneously: 1) align representations of rotated views from identical images; 2) contrast them against dissimilar images; and 3) maintain semantic consistency between masked and unmasked branches. Our implementation incorporates a dynamic dictionary and momentum update mechanism following the MoCo framework.

First, we exploit the extracted features $\{\mathbf{z}_i\}_{i=1}^4$ from the input image and three randomly rotated images in the unmasked teacher branch to compute the rotation-averaged features. The global average pooling operation is used to obtain their global representations, which are projected to a new feature to prevent confusion with other tasks via an MLP projector. Then the rotation-averaged feature $\bar{\mathbf{z}}$ is obtained by averaging the features with four rotation angles, i.e.,

$$\bar{\mathbf{z}} = \frac{1}{4}\sum_k \mathbf{z}_k \tag{5}$$

where $\mathbf{z}_k$ is the projected multiangle features from the unmasked teacher branch.

The extracted masked multiangle features $\{x_k^{\text{mask}}\}_{i=1}^4$ in the student branch are projected by an MLP projector for CL. We exploit infoNCE loss [34] to align the rotation-averaged features with the masked features with different rotation angles by discriminating them from other images

$$L_{\text{contrast}} = -\sum_{k=1}^{4} \log \frac{\exp\left(\text{sim}\left(\bar{\mathbf{z}}, \mathbf{z}_k^{\text{mask}}\right)/\tau\right)}{\sum_j \exp\left(\text{sim}\left(\bar{\mathbf{z}}, \mathbf{z}_j\right)/\tau\right)} \tag{6}$$

where sim(·) is cosine similarity, τ is a temperature parameter, and j indexes all features in the batch and queues. The rotation-averaged features from the teacher branch are saved in the queue and updated in each iteration.

The averaging operation inherently suppresses rotation-specific patterns through the fusion of multiangle features, thereby disentangling the rotation transformations from semantic content. In CL, the averaged features serve as stabilized anchors to which features from different rotation angles are attracted. This design reduces the variance among rotation-variant features while compelling the model to concentrate on rotation-invariant essential characteristics.

D. Masked Local Semantic Alignment

While MIM learns contextual information by reconstructing randomly masked input images, this approach presents challenges for remote sensing data. The dense distribution of small objects in such imagery means that random masking often results in critical small-target information loss, compromising reconstruction quality.

We employ masked local semantic alignment by minimizing the distribution difference between the masked features from the student model and the augmented features from the teacher model. This alignment enforces semantic consistency between masked regions and augmented views through prototype matching of position-corresponding pairs. The teacher model's features are generated via random cropping from original images, where we utilize the overlapping regions between cropped and masked images as CMID [21].

First, we utilize the position information to match the cropped patches by calculating the Euclidean distance between the cropped features and masked features using their absolute position. Then we select the top K features pairs with smallest distance as the position-matched pairs $\{\mathbf{z}_i^l, \hat{\mathbf{z}}_i^l)\}_{i=1}^K$ for subsequent calculation. In order to avoid semantic confusion, we project the matched pairs to a new feature space and obtain N local semantic prototypes $\mathbf{P} \in \mathbb{R}^{N\times d}$ which could be updated by the training process. The similarity distributions $\mathbf{p}_i$ ($\hat{\mathbf{p}}_i$) between $\mathbf{z}_i^l$ ($\hat{\mathbf{z}}_i^l$) and semantic prototypes $\mathbf{P}$ can be calculated by

$$\mathbf{p}_i = \text{Softmax}\left(\frac{1}{\tau_s}\text{sim}\left(\mathbf{z}_i^l, \mathbf{P}\right)\right) \tag{7}$$

$$\hat{\mathbf{p}}_i = \text{Softmax}\left(\frac{1}{\tau_t}\text{sim}\left(\hat{\mathbf{z}}_i^l \mathbf{P}\right)\right) \tag{8}$$

where sim(·) represents the cosine similarity, and τ_s, τ_t is temperature parameters. We set τ_s and τ_t as 0.2 and 0.07, respectively.

Finally, the cross-entropy loss is used to implicitly model the spatial dependence between ground objects and enhance the spatial coherence of features, i.e.,

$$L_{\text{local}} = -\frac{1}{K}\sum_{i=1}^{K} \mathbf{p}_i \log\left(\hat{\mathbf{p}}_i\right). \tag{9}$$

Masked local semantic alignment forces the model to learn pixel-level or region-level semantic consistency by comparing the corresponding local features between the masked region and the cropped image, and infers the local semantics of the masked branch through the contextual information from

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unmasked patches, thus enhancing the robustness to some missing information.

E. Synergy of Multitask Learning

Our framework jointly optimizes three synergistic tasks through a combined loss

$$L_{\text{total}} = L_{\text{reconstruct}} + L_{\text{Local}} + L_{\text{contrast}}. \tag{10}$$

The proposed framework establishes a hierarchical feature learning system through three coordinated tasks: 1) masked rotation-averaged CL suppresses rotational variance by enforcing high-level semantic invariance; 2) masked multiangle reconstruction recovers fine-grained textural details and enforces low-level rotation-equivariant constraints; and 3) masked local semantic alignment resolves regional incompleteness via patch-level correspondence learning. Specifically, local alignment ensures precise feature matching to prevent global pooling ambiguities and enhances reconstruction accuracy by maintaining detail consistency, while CL's feature-space alignment prevents semantic drift in multiangle reconstructions. This integrated approach simultaneously achieves rotational invariance, fine-grained semantic optimization, and enhanced spatial reasoning for comprehensive remote sensing image understanding.

IV. Experiment

In this section, we first detail the experimental setups for the pretraining stage. Subsequently, we fine-tune the pretrained HRI model on various remote sensing tasks. Finally, we conduct ablation studies and further analyses.

A. Experimental Setting for Pretraining

1) Dataset: Our HRI is pretrained on the MillionAID dataset [36], a large-scale benchmark for remote sensing scene classification containing 51 distinct scene classes. MillionAID comprises 1 000 848 images captured by multiple remote sensing sensors, offering global coverage, high spatial resolution (0.5–11.4 m), and RGB spectral bands. For pretraining, we utilize 990 848 images (the designated test subset) resized to 224×224 pixels as network inputs.

2) Implementation Details: We use Swin-Base [37] and P2T-Base [38] as the encoder. During training, we use the Adam optimizer with a base learning rate of 0.000375, weight decay of 0.02 for Swin-Base, and the AdamW optimizer with a base learning rate of 0.0001, weight decay of 0.05 for P2T-Base. The batch size is set to 48. We adopt a cosine learning rate scheduler with 15 warm-up epochs. The mask ratio is set as 0.6 based on the results of the ablation study. For momentum-based CL, we initialize the EMA momentum rate to 0.996, progressively increasing it to 1 via cosine annealing. The infoNCE loss parameters follow MoCo [22], with temperature 0.2 and queue size 66 048. The number of position-matched pairs K and prototypes N in the masked local semantic alignment task are set to 20 and 2048, respectively, the same setting as [21]. Pretraining is performed on four NVIDIA 3090 GPUs under the PyTorch framework.

TABLE II

Classification Performance Comparison (%) of SSL Methods on the UCM Testing Set. The Best and the Second-Best Results Are Marked in **Bold** and Underline, Respectively. † Indicates the Method Implemented by Ourselves

Method	Backbone	Pretrain Epoch	Finetune (Train/Test)			Linear Probe (Train/Test)		
			2:8	5:5	8:2	2:8	5:5	8:2
MoCo-v2† [34]	ResNet-50	800	91.05	96.29	98.44	92.01	95.21	98.44
MoCo-v3† [34]	ViT-B	300	95.43	97.27	99.48	92.01	95.21	98.18
MAE† [22]	ViT-B-RVSA	1600	92.25	96.97	98.70	92.37	96.00	97.86
MAE† [22]	ViTAE-B-RVSA	1600	93.51	97.17	99.22	72.80	85.24	89.29
CMID† [21]	ResNet-50	200	94.65	97.85	99.22	94.47	97.66	98.44
CMID† [21]	Swin-B	200	95.13	97.95	<u>99.74</u>	92.37	95.70	96.88
HRI (Ours)	Swin-B	50	<u>96.45</u>	<u>98.14</u>	**100.00**	<u>94.53</u>	<u>96.68</u>	**98.96**
HRI (Ours)	P2T-B	50	**98.32**	**98.63**	<u>99.74</u>	**95.43**	**96.88**	<u>98.70</u>

B. Results of Scene Classification

Scene classification is a fundamental task to evaluate the performance of backbone networks, aiming to assign a class label to a remote sensing image.

1) UC Merced Land Use Dataset: The UC Merced Land Use (UCM) dataset comprises 21 scene classes, with 100 images per class (256×256 pixels, 0.3-m resolution). These images are manually extracted from the USGS National Map Urban Area Imagery collection, which covers diverse urban regions across the United States.

2) Experimental Setup: We randomly split the UCM dataset into training and testing sets using different ratios (2:8, 5:5, 8:2). The model is trained for 300 epochs using the AdamW optimizer with a cosine learning rate scheduler, a batch size of 64, a base learning rate of $2.5e^{-4}$, and a weight decay of 0.05. A 10-epoch warmup is applied. For linear probing, we use a learning rate of 0.025. Data augmentation follows the settings in [23]. During fine-tuning, both the backbone and classification head are updated, whereas only the classification head is trained in the linear probing setting. We evaluate performance using overall accuracy (OA). All baseline methods are implemented using their publicly available codes and default parameter settings.

3) Results: The classification results are presented in Table II. Our method consistently outperforms baselines across all data splits (2:8, 5:5, 8:2), whether through full model fine-tuning or linear probing. When fine-tuning the full model, With P2T-Base backbone, our method surpasses ViT-B-RVSA by 6.07%, 1.66%, and 1.04%, and ViTAE-B-RVSA by 4.81%, 1.46%, and 0.52% (versus MAE-pretrained [23] at 1600 epochs) for 2:8, 5:5, and 8:2 splits, respectively. With Swin-Base, HRI achieves gains of 4.20%, 1.17%, and 1.30% over ViT-B-RVSA (MAE-pretrained) and 2.94%, 0.97%, and 0.78% over ViTAE-B-RVSA. Compared to CMID [21] (same Swin-B backbone), our method improves by 1.32%, 0.19%, and 0.26%; with P2T-B [38], HRI excels by 3.19% and 0.68% for 2:8 and 5:5 splits. For linear probing tasks, using Swin-Base, our method outperforms CMID (Swin-B) by 2.16%, 0.98%, and 2.08%; with P2T-Base, the gains are 3.06%, 1.18%, and 1.82% for the respective splits. Even against CMID (ResNet-50), which typically benefits from CNN biases in linear probing, our method achieves significant improvements across most splits except for a marginal lag in the 5:5 split. HRI with P2T-Base exceeds that with Swin-Base backbone under 2:8 and 5:5 split, while the performance between HRI

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TABLE III
FLOPS AND PARAMETER SIZES OF DIFFERENT BACKBONES ARE REPORTED. ALL INPUTS ARE RESIZED TO (1, 3, 224, 224), AND THE RESULTS ARE EVALUATED ON AN NVIDIA 3090 GPU USING THE THOP LIBRARY

Backbone	Performance (%)	FLOPs (G)	Params. (M)	FPS
ViT-B	95.43	16.862	85.663	167.08
ViT-B-RVSA	92.25	16.779	86.004	39.91
ViTAE-B-RVSA	93.51	13.166	67.545	35.27
Swin-B (Ours)	96.45	5.304	30.804	42.68
P2T-Base (Ours)	98.32	6.312	35.565	37.68

with Swin-Base and P2T-Base is reduced when increasing the labeled samples and exceeds HRI with P2T-Base under 8:2 split.

In addition, we report the training cost and inference speed to highlight the efficiency of our model. As shown in Table III, our pretrained backbone achieves superior classification performance with lower computational complexity and fewer parameters compared to the transformer-based competitors, i.e., ViT-RVSA and ViTAE-B-RVSA [23].

C. Results of Object Detection

1) Dataset: The DOTA dataset [39] is a large-scale benchmark for oriented object detection in aerial imagery, containing 2806 high-resolution images (both RGB and grayscale) with pixel dimensions ranging from 800 × 800 to 4000 × 4.000. It is split into 1411 training, 458 validation, and 937 testing images, annotated with 188 282 instances across 15 object categories. Each instance is labeled with an arbitrary quadrilateral to capture objects of diverse scales, orientations, and shapes. We train the model using the data from the training and validation sets. Following [40], we preprocess the dataset by cropping images into 1024 × 1.024-pixel patches with a stride of 824 pixels to balance computational efficiency and context preservation.

2) Experimental Setting: We employ the ORCN framework [41] for oriented object detection, consistent with CMID [21], to ensure fair comparison. For P2T-Base and Swin-Base backbones, we use the AdamW optimizer with a learning rate of $1e^{-4}$ and weight decay of 0.05. Other competitors follow their official hyperparameters. The batch size is set as 2, and all models are trained for 12 epochs with learning rate reductions by 10 after the 8th and 11th epochs. Evaluation is performed using mean average precision (mAP) via the official DOTA test server.

3) Results: The object detection results (Table IV) demonstrate our method's strong performance, achieving 77.83% and 78.07% mAP with Swin-Base and P2T-Base backbones, respectively. Using Swin-Base, our approach outperforms CMID (200-epoch pretrained) by 0.54%, while showing comparable performance to ViT-B-RVSA and ViTAE-B-RVSA, which are pretrained for 1600 epochs. With P2T-Base, we surpass Swin-Base CMID by 0.78%, and exceed ViT-B-RVSA and ViTAE-B-RVSA by 0.23% and 0.36%, respectively. Visualization results (Fig. 3) further confirm our method's superior target identification accuracy and higher classification confidence compared to baseline CMID.

D. Results of Change Detection

RS Change Detection detects changes by comparing images of the same geographic area acquired at different times.

1) Dataset: The CDD dataset consists of three types: synthetic images with no relative object movement, synthetic images with small relative object movement, and real remote sensing images with seasonal variations (acquired from Google Earth). The seasonal remote sensing images include 16 000 image sets, with a resolution of 256 × 256 pixels (10 000 for the training set and 3000 for testing and validation), and a spatial resolution ranging from 3 to 100 cm.

2) Experimental Setting: Following [21], we adopt the BiT framework for evaluation on the change detection task. We use the AdamW optimizer with a learning rate of $1.2e^{-4}$ and a weight decay of 0.01, and set the batch size to 8. All models are trained for 200 epochs, and the learning rate decreases linearly during the training process. We report the mean $F1$-score (m$F1$) on the test set.

3) Results: As shown in Table V, our HRI obtains the best performance. When using P2T-Base as the feature extractor, our method achieves the highest score with 97.18%. The performance with Swin-Base is slightly decreased but still exceeds most methods. Specifically, compared with the pretrained models in remote sensing, our model excels CMID with Swin-Base as backbone by 1.95% and 2.41% under Swin-Base and P2T-Base, respectively. When compared with the models pretrained by MAE, our method with P2T-Base is 5.20% and 1.36% higher than ViT-B-RVSA and ViTAE-B-RVSA. As shown in Fig. 4, our method could recognize the edge information and road information.

E. Results of RS Semantic Segmentation

1) Dataset: We evaluate semantic segmentation performance on two ISPRS benchmark datasets: Potsdam and Vaihingen. The Potsdam dataset captures a dense urban environment with complex architecture, containing 28 ultrahigh-resolution images (6000 × 6000 pixels, 5-cm GSD). In contrast, the Vaihingen dataset represents a rural village landscape with 33 images averaging 2494 × 2.064 pixels at 9-cm GSD. Both datasets share six land cover categories: impermeable surfaces, buildings, low vegetation, trees, cars, and background. Following CMID [21] and [43], we preprocess the data by extracting 512 × 512 pixel patches with a 256-pixel stride and use their default train-test splits.

2) Experimental Setting: Following CMID [21], we employ the UperNet framework [44] with various SSL pretrained backbones. The models are fine-tuned for 50 epochs (Potsdam) and 100 epochs (Vaihingen) using AdamW optimizer (lr = $6e^{-5}$, weight decay = 0.05) with cosine annealing scheduling. We maintain a consistent batch size of eight across both datasets to ensure fair comparison.

3) Results: Semantic segmentation results (Table VI) demonstrate consistent improvements across both datasets. Using Swin-Base, our method outperforms CMID [21] by 0.90% (Potsdam) and 1.41% (Vaihingen) in mIoU, while P2T-Base achieves gains of 0.57% and 1.17%, respectively.

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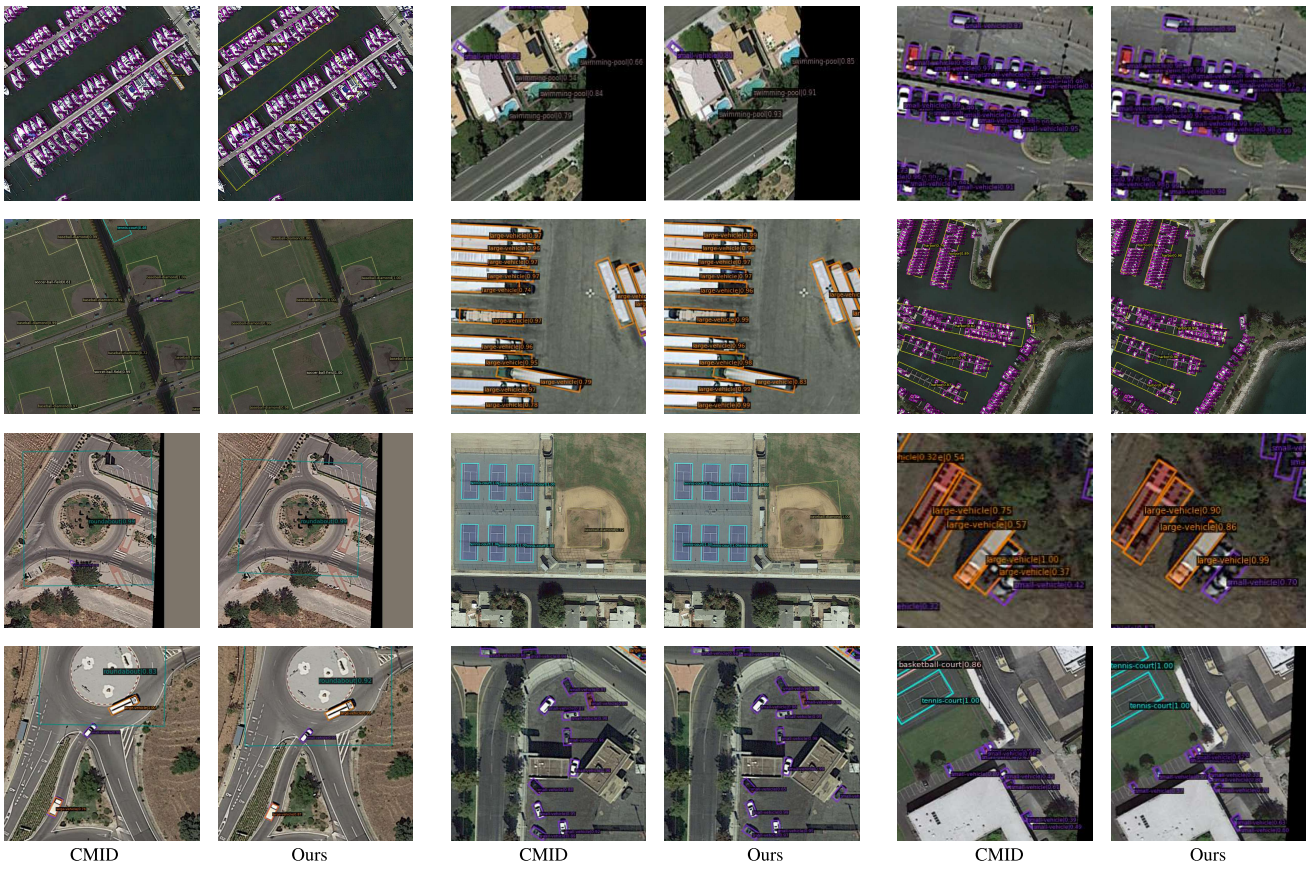


Fig. 3. Object detection visualization of CMID and our HRI on the DOTA test set.

TABLE IV

OBJECT DETECTION PERFORMANCE (%) OF VARIOUS SSL METHODS ON THE DOTA TEST SET. THE BEST AND THE SECOND-BEST RESULTS ARE MARKED IN **Bold** AND UNDERLINE, RESPECTIVELY. † INDICATES THE METHOD IMPLEMENTED BY OURSELVES

Method	Backbone	Dataset	Epoch	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
MoCo-v2 [34]	ResNet-50	ImageNet	200	89.27	78.29	51.21	71.67	78.40	81.32	87.92	90.89	84.34	84.88	56.30	60.91	67.39	65.32	50.72	73.25
MAE† [22]	ViT-B-RVSA	MillionAID	1600	89.17	82.71	**55.38**	73.69	79.13	84.99	87.96	90.87	**87.62**	**86.55**	58.95	69.40	76.98	75.83	68.29	77.84
MAE† [22]	ViTAE-B-RVSA	MillionAID	1600	89.36	81.94	55.24	71.62	79.22	**85.19**	**88.00**	**90.90**	87.23	86.36	62.21	70.31	76.99	**77.97**	63.13	77.71
CMID† [21]	ResNet-50	MillionAID	200	89.28	83.09	51.07	72.22	76.55	81.06	87.46	90.87	87.42	85.31	64.22	63.78	65.64	66.44	59.02	74.90
CMID† [21]	Swin-B	MillionAID	200	89.57	83.45	53.26	75.39	78.92	83.83	87.58	90.89	87.56	86.00	65.14	68.53	73.75	72.32	63.17	77.29
HRI (Ours)	Swin-B	MillionAID	50	89.64	**84.59**	54.14	**76.70**	79.27	84.86	87.59	90.89	87.35	85.72	**66.83**	67.51	**77.00**	72.18	64.20	77.83
HRI (Ours)	P2T-B	MillionAID	50	**89.85**	84.15	55.29	72.29	**79.59**	85.08	87.75	90.89	87.38	85.85	64.91	**72.47**	76.95	69.27	**69.34**	**78.07**

TABLE V

CHANGE DETECTION PERFORMANCE (%) OF VARIOUS SSL METHODS ON THE CDD TEST SET. THE BEST AND THE SECOND-BEST RESULTS ARE MARKED IN **Bold** AND UNDERLINE, RESPECTIVELY. † INDICATES THE METHOD IMPLEMENTED BY OURSELVES

Method	Backbone	Pre-train Dataset	Pre-train Epoch	mF1
MoCo-v2† [34]	ResNet-50	ImageNet	800	96.43
MoCo-v3† [42]	ViT-Base	ImageNet	300	91.00
MAE† [22]	ViT-B-RVSA	MillionAID	1600	91.98
MAE† [22]	ViTAE-B-RVSA	MillionAID	1600	95.82
CMID† [21]	ResNet-50	MillionAID	200	94.40
CMID† [21]	Swin-B	MillionAID	200	94.77
HRI (Ours)	Swin-B	MillionAID	50	96.72
HRI (Ours)	P2T	MillionAID	50	**97.18**

TABLE VI

SEMANTIC SEGMENTATION PERFORMANCE (%) OF VARIOUS SSL METHODS ON BOTH POTADAM AND VAIHINGEN DATASETS. THE BEST AND THE SECOND-BEST RESULTS ARE MARKED IN **Bold** AND UNDERLINE, RESPECTIVELY. † INDICATES THE METHOD IMPLEMENTED BY OURSELVES

Method	Backbone	Pre-train Epoch	Potsdam		Vaihingen	
			OA	mIoU	OA	mIoU
MoCo-v2 [34]	ResNet-50	200	92.74	87.02	90.26	79.16
MAE [22]	ViT-B-RVSA	1600	92.40	86.37	89.54	77.29
MAE [22]	ViTAE-B-RVSA	1600	92.62	86.61	89.79	78.17
CMID† [21]	ResNet-50	200	92,12	85.90	89.79	77.37
CMID† [21]	Swin-B	200	93.29	87.89	90.95	80.80
HRI (Ours)	Swin-B	50	**93.74**	**88.78**	**91.53**	**82.21**
HRI (Ours)	P2T	50	93.56	88.46	91.44	81.97

Compared to CMID with ResNet-50, we observe more substantial improvements of 2.88% and 4.84% with the Swin-Base backbone in mIoU on each dataset. Visualization results (Fig. 5) confirm our method's superior performance in detecting small objects and capturing precise edge information compared to CMID.

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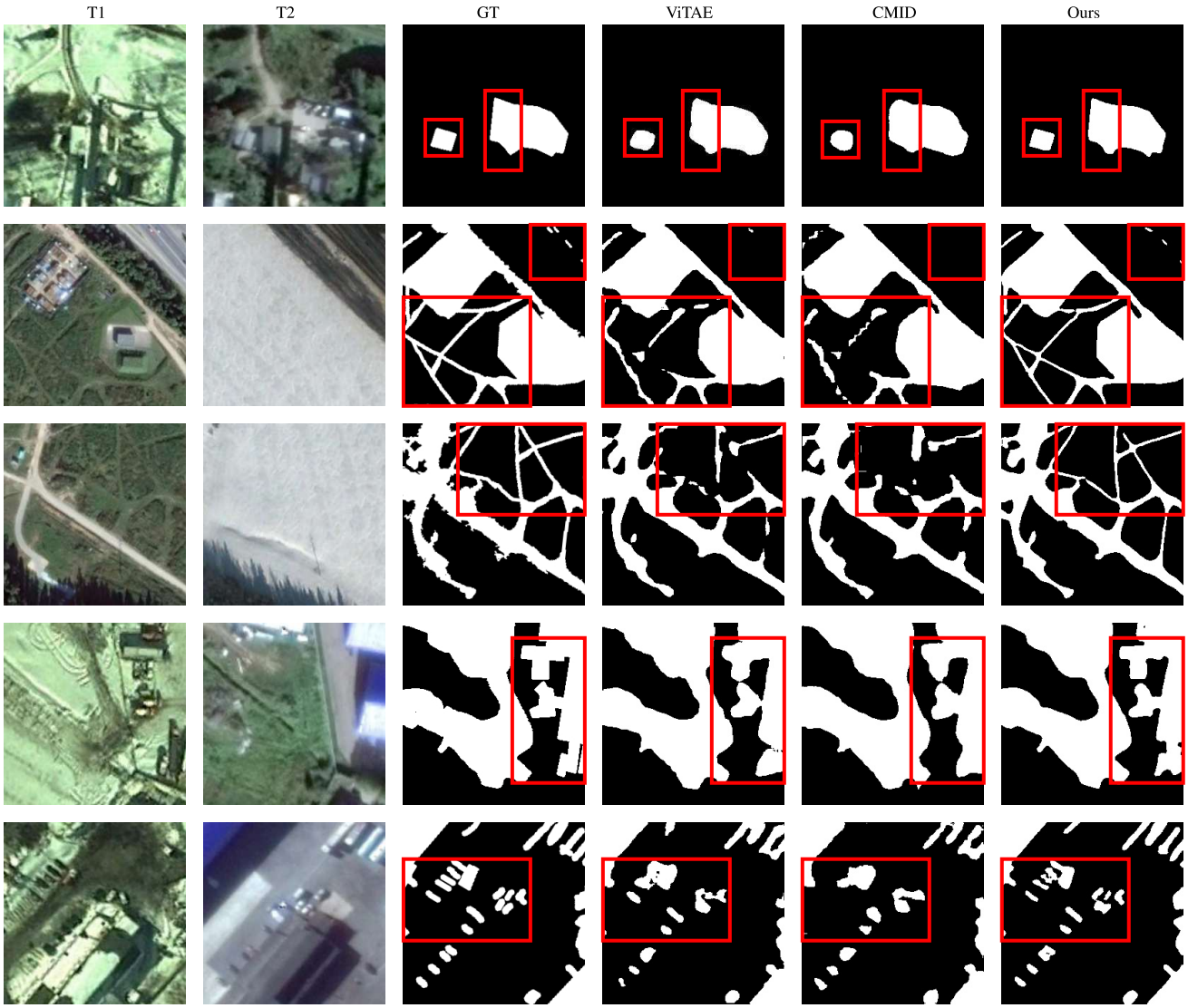


Fig. 4. Change detection visualization of two different competitors ViTAE and CMID, and our HRI with P2T-Base backbone on the CDD dataset.

TABLE VII
ABLATION STUDY RESULTS OF OUR PROPOSED METHOD ON UCM CLASSIFICATION AND POTSDAM SEGMENTATION DATASET

MLSA.	MMR.	MRCL.	8:2 Cls (Acc %)	Seg (mIoU %)
✓		✓	98.70	87.86
	✓	✓	97.40	88.06
✓	✓		97.92	88.33
✓	✓	✓	98.70	88.46

F. Ablation Studies and Further Analysis

1) Ablation Studies:

a) Effects of proposed modules: Ablation study (Table VII) investigates the contribution of each proposed module on linear probing classification and segmentation downstream tasks with P2T-Base as backbone. "MLSA" refers to masked local semantic alignment, "MMR" is masked multiangle reconstruction, and "MRCL" is masked rotation-averaged CL. When removing "MMR," we exploit traditional masked image reconstruction to perceive the textual information. When removing "MRCL," we only utilize the same MoCo CL method as [21] without considering the features from different rotation angles.

From the results, we observe that distinct contributions from each proposed module. Removing MMR causes a 0.60% segmentation performance drop, confirming its importance in learning rotation-invariant low-level features for pixel-level understanding. The MLSA task proves crucial for classification (1.30% drop when removed) and moderately benefits segmentation (0.30% drop), demonstrating its effectiveness in local semantic alignment. MRCL removal leads to a 0.78% classification and 0.13% segmentation performance decrease, verifying its role in capturing robust rotation-invariant features through noise-reducing feature averaging. We start by visualizing the extracted representations on the UCM dataset using the t-SNE [45] method. As shown in Fig. 6, the representations learned by the model pretrained simultaneously on all three tasks exhibit greater interclass dispersion and intraclass compactness than those obtained from models pretrained with any

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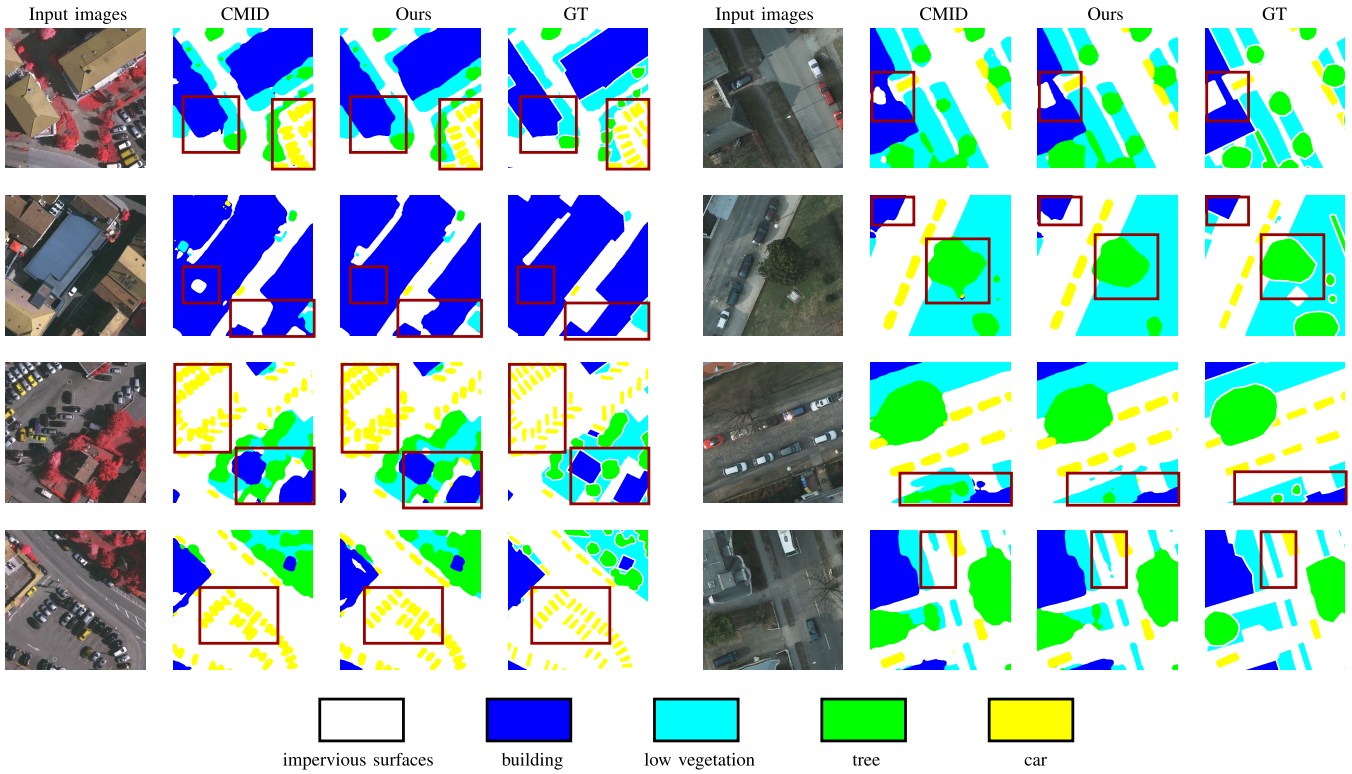


Fig. 5. Segmentation visualization results of CMID and our HRI on both Potsdam and Vaigingen datasets.

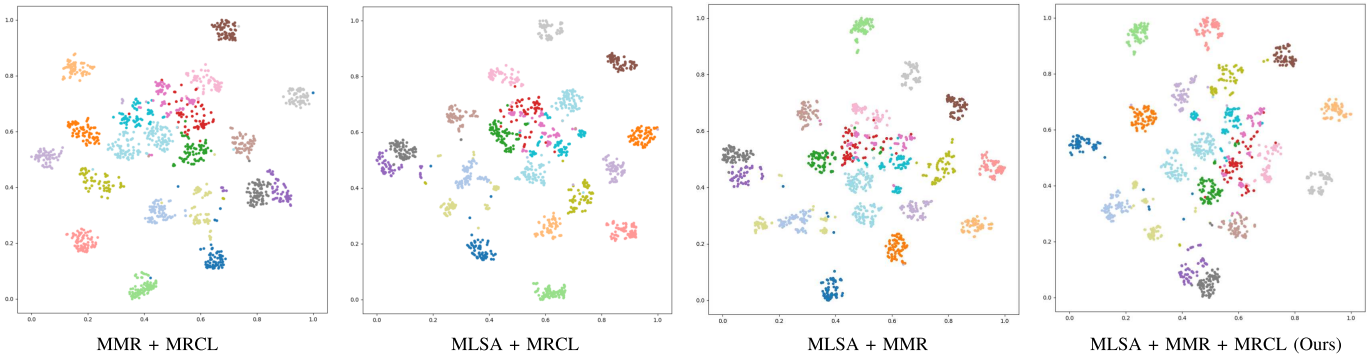


Fig. 6. t-SNE visualization results of the ablation study.

TABLE VIII

PERFORMANCE (%) OF DIFFERENT MASK RATIOS IN MMR TASK ON UCM CLASSIFICATION AND POTSDAM SEGMENTATION DATASET

Mask Ratio (r)	2:8 Cls (Acc %)	Seg (mIoU %)
0.4	98.08	87.97
0.6	98.32	88.46
0.8	97.66	88.17

single task omitted. This suggests that the multitask framework encourages the learning of more discriminative and robust feature representations.

b) Effects of mask ratios: We investigate the impact of mask ratios r in our masked multiangle reconstruction module, using P2T-Base as the feature extractor while keeping other parameters fixed. As shown in Table VIII, both full fine-tuning classification (2:8 split) and segmentation (Potsdam) performance degrade with extreme mask ratios. Excessively high ratios (e.g., $r > 0.8$) obscure critical structural features, disrupting semantic coherence and amplifying noise across multiangle reconstructions. Conversely, overly low ratios (e.g., $r < 0.4$) promote overfitting to local features while neglecting global semantic relationships, trapping the model in suboptimal solutions. Our experiments demonstrate that a balanced 0.6 mask ratio optimally preserves both local detail and global context for effective pretraining. We provide the t-SNE visualization results of the different mask ratios in the MMR module in Fig. 7, and the clusters with a mask ratio of 0.6 are better interclass separated and more intraclass gathered than others.

2) Further Analysis:

a) Visualization of masked multiangle reconstruction: To demonstrate the effectiveness of our proposed self-supervised framework, we provide some reconstruction results in Fig. 8.

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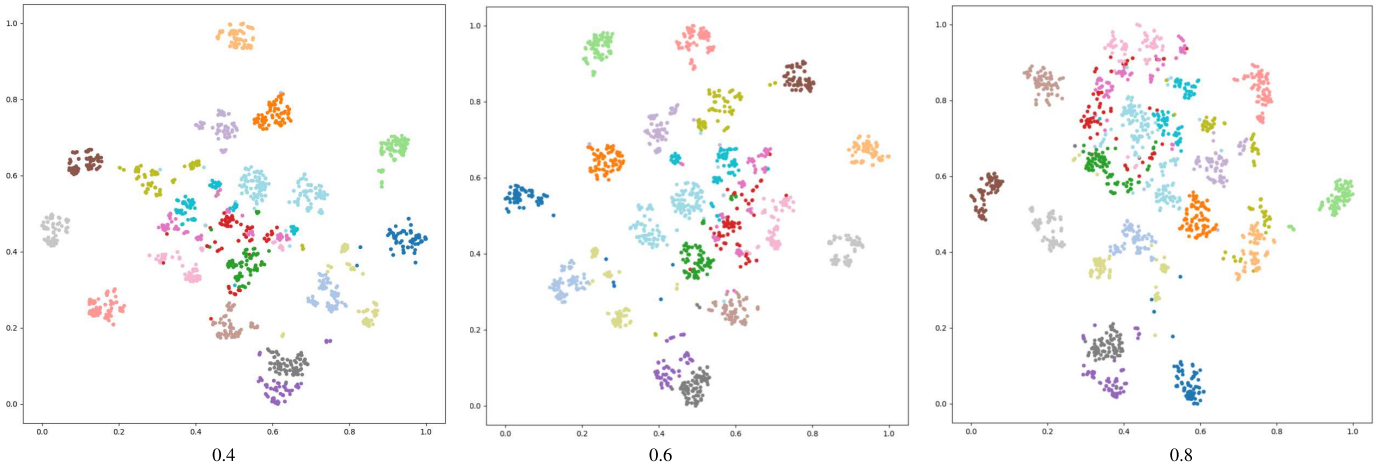


Fig. 7. t-SNE visualization results of the different mask ratios in the MMR module. We use a 0.6 mask ratio in the experiment.

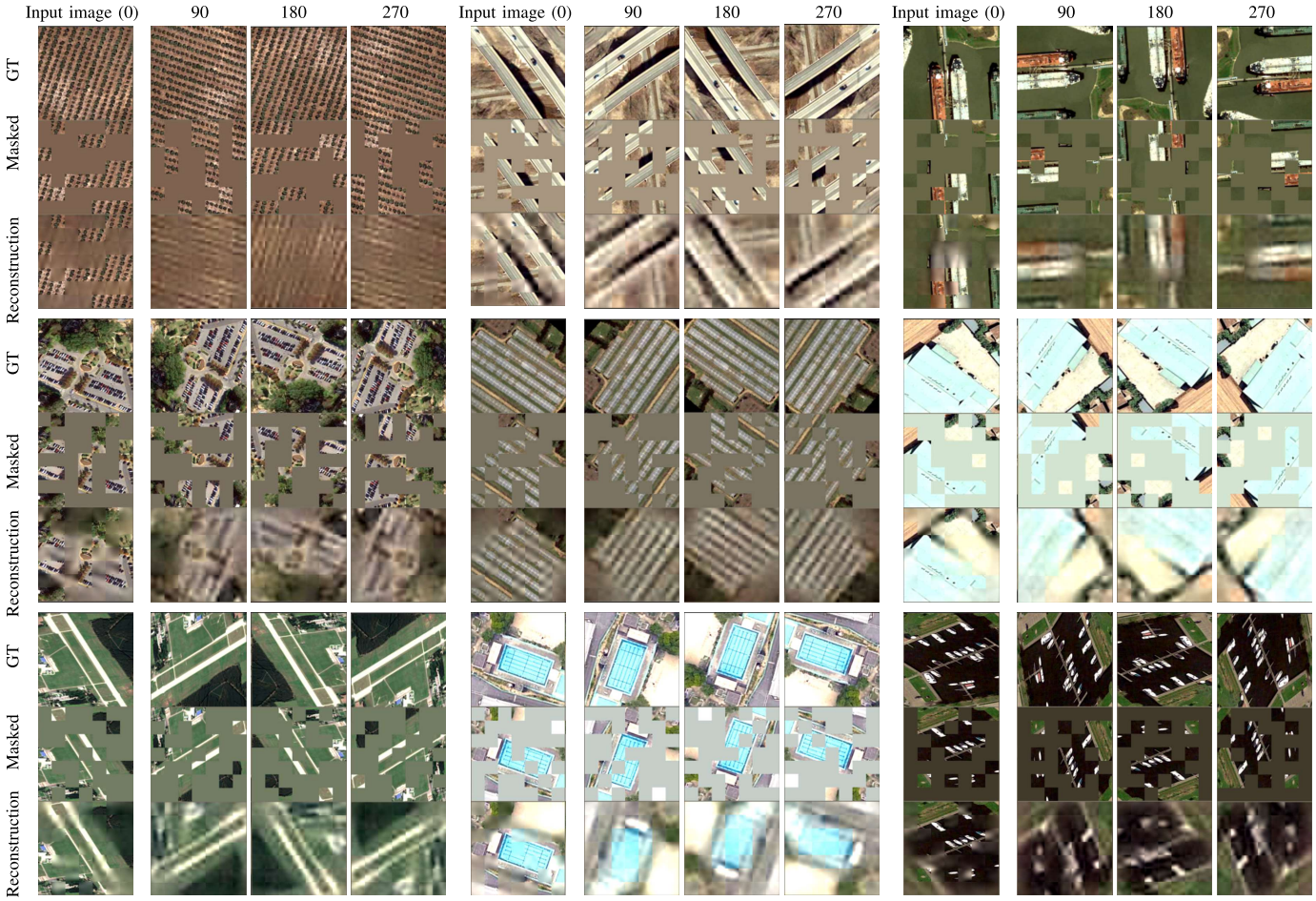


Fig. 8. Visualization of reconstruction results from our HRI method across various rotated views.

Each input image is arranged in three rows. The first row displays the ground-truth images, which serve as reconstruction targets for model pretraining. The second row shows the masked images: only the first image is the original masked input for reconstruction, while the remaining images are generated by rotating this first masked image by the corresponding angles.

The reconstructed images demonstrate that the model effectively perceives directional and textual information and successfully reconstructs them to different rotated orientations correctly, achieving multiangle image reconstruction from a single-angle input. The successful multiangle reconstructions confirm that our joint self-supervised training framework effectively learns rotation-invariant feature representations, enabling robust target perception under arbitrary rotations.

b) Robustness to rotation: To rigorously validate the robustness to rotation of our pretrained model, we employ linear probing classification for evaluation with Swin-Base

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TABLE IX
CLASSIFICATION PERFORMANCE (%) OF THE RANDOM ROTATED IMAGES ON THE UCM DATASET

Method	CMID (Train:Test)			HRI (Train:Test)		
	2:8	5:5	8:2	2:8	5:5	8:2
Without Rotation	92.37	95.70	96.88	94.53	96.68	98.96
With Rotation	89.42	94.14	94.53	93.15	96.19	97.66
Performance Gap Δ	-2.95	-1.56	-2.35	-1.38	-0.49	-1.30

as a backbone. This approach isolates and quantifies the effectiveness of the pretraining framework, as it ensures that performance improvements are attributed solely to the learned rotation-invariant representations rather than fine-tuning adjustments. We conduct an experiment on the UCM dataset and the results are shown in Table IX. With Rotation represents that we apply random rotations to input images during inference. Experimental results demonstrate that our model significantly mitigates performance degradation induced by rotational variations. This indicates that the proposed framework effectively learns rotation-invariant feature representations through its self-supervised pretraining paradigm, which explicitly disentangles orientation-dependent and orientation-agnostic visual patterns. Consequently, the model exhibits enhanced robustness to arbitrary rotations, generalizing well to unseen orientations without requiring explicit geometric augmentation during training.

V. DISCUSSION

While our pretraining framework demonstrates promising performance across the downstream tasks, it is crucial to acknowledge and discuss its inherent limitations, particularly concerning the unique characteristics of remote sensing imagery.

A primary limitation stems from our masking strategy during the pretraining phase. Our approach, which employs a patch-based random masking protocol with a relatively large patch size, may be suboptimal for remote sensing scenes. These scenes are often characterized by a high density of small and micro-objects (e.g., vehicles, ships, aircraft). When a masked patch encompasses multiple such small objects or intersects the boundary between an object and its background, the model is forced to reconstruct or learn from a context that is inherently corrupted and imprecise. This leads to the learning of coarse and distorted feature representations for these critical regions as shown in Fig. 8.

Consequently, this pretraining paradigm can result in the loss of fine-grained information pertaining to densely packed, small targets. The model, therefore, may fail to develop effective feature representations for these elements. This shortcoming directly translates to compromised performance in downstream tasks that are highly sensitive to such details, most notably in small object detection, where the accuracy of identifying and localizing minuscule objects is paramount.

To address this issue, more sophisticated masking strategies tailored to remote sensing imagery should be explored. Rather than relying on random patch masking, object-aware masking approaches could be devised. By incorporating saliency maps or low-level segmentation cues, these approaches could strategically preserve small objects during masking, enabling the model to learn more robust and fine-grained representations from such critical regions.

VI. CONCLUSION

We propose HRI learning, a self-supervised framework that jointly optimizes masked multiangle reconstruction, masked local semantic alignment, and masked rotation-averaged CL to address rotational variability in remote sensing imagery. By explicitly disentangling content and orientation through multitask pretraining, HRI achieves state-of-the-art performance on classification, rotated object detection, segmentation, and change detection tasks. Extensive experiments demonstrate its superiority over existing methods, with ablation studies validating the necessity of each component. The framework's robustness to arbitrary rotations and occlusions makes it particularly suitable for large-scale geospatial analysis. Future work will explore multimodal extensions and dynamic masking strategies for complex scenes.

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